

Proposal: Dubjly - AI-Powered Video Dubbing and Translation Platform

Problem Statement:

Many students struggle to develop a deep understanding of the English language, which limits their access to high-quality educational resources. Since most academic content is in English, this creates a significant learning barrier, making it challenging for students to comprehend complex material. Additionally, finding educational resources in Arabic that match the quality of foreign sources is rare, further exacerbating the issue.

Solution Overview:

Dubjly is an AI-powered platform designed to enhance students' learning experiences by:

- Dubbing videos with natural, native Arabic voices generated by AI.
- Providing an interactive chatbot that explains content, answers questions, and facilitates discussions about the video.
- Generating summaries of videos.
- Highlighting key terms and defining them in their original English without dubbing.

By integrating machine translation, dubbing, and AI-powered interaction, Dabjali creates an immersive learning experience that helps students grasp complex concepts without language barriers.

Target audience

- School and university students (non-native English speakers)
- Universities and online learning platforms (Coursera, Udemy) that need local content.
- Researchers, business professionals, and tech enthusiasts needing translated educational content.

2. Features and Components

2.1 Core Features

Video Translation & Dubbing:

- Users can upload a video or provide a link.
- AI extracts audio, translates it into Arabic, and generates a dubbed version using advanced TTS (Text-to-Speech) models.
- English scientific terms are preserved and excluded from dubbing.

Chatbot Assistance:

- An AI-powered chatbot that provides explanations, Q&As, and video summaries.
- Users can ask specific questions and receive detailed answers related to the video.

Terminology Learning Panel:

- Important technical terms remain in English and are highlighted in a dedicated panel.
- Clicking on a term provides an instant Arabic definition and example usage.

Summarization Feature:

- Generates concise summaries of videos, allowing users to quickly grasp key points.

Interactive Learning Enhancements:

- Users can engage with quizzes based on the video's content.
- Ability to save key terms and revisit them later for learning.

3. Technical Architecture

3.1 System Overview

The system follows a microservices architecture, ensuring modularity and scalability.

It consists of:

- Frontend (UI/UX): Built with Dajango or python streamlit .
- Backend API: Developed using FastAPI (Python) .

- **Database:** PostgreSQL or MongoDB for storing user data and video metadata.
 - **AI Services:**
 - Speech-to-Text (STT): OpenAI Whisper (Large-v3) or AssemblyAI
 - Identify Scientific Terms : BERT/Roberta
 - Machine Translation (MT): Fine-tuned LLAMA or GPT-based model or DeepL API or Google Cloud Translation API
 - Text-to-Speech (TTS): Narakeet or ElevenLabs
 - Chatbot AI: LLM with RAG (Retrieval-Augmented Generation) , GroqAPI
 - **Cloud Storage:** AWS S3 or Google Cloud Storage for video files
 - **Task Queue:** Celery with Redis for background processing
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4. Technical Components

4.1 Frontend (UI/UX)

Technology Stack: Django

Features :

- User authentication(django-allauth or django-rest-framework-jwt)
- Video Upload & Processing Dashboard (django-resumable.js)
- Interactive Video Player with Dual Subtitles(Video.js or Plyr.js)
- AI Chatbot Interface (LangChain + FastAPI for chatbot API)
- Terminology Pane

4.2 Backend API

Technology Stack: RESTful or GraphQL API to handle authentication, video processing, chatbot integration, and terminology management.

Endpoints:

- POST /api/auth/login/ → User Authentication
- POST /api/videos/upload/ → Uploads video
- POST /api/videos/process/ → Trigger video processing
- GET /api/videos/{id}/ → Get video details & subtitles
- POST /api/chatbot/query/ → Get chatbot responses
- GET /api/terminology/ → Fetch terminology & definitions

4.3 Database : PostgreSQL (recommended) or MySQL

Schema:

- **Users Model** (Authentication & Preferences)
- **Videos Model** (Metadata & Processing Status)
- **Terminologies Model** (Key Terms & Definitions)
- **Chatbot Interactions Model** (User Queries & AI Responses)

4.4 AI Processing Pipelines

1- Audio Extration :

- Extract audio clearly from videos without any external effects.

2-Speech-to-Text (STT):

- Whisper API transcribes English video audio to text.

3-Machine Translation (MT):

- Fine-tuned LLaMA or GPT-based Arabic translation.

Text-to-Speech (TTS):

- ElevenLabs/Coqui AI generates human-like Arabic audio.

Chatbot AI (RAG-based)

- Uses a vector database (e.g., Pinecone, FAISS) to fetch relevant context for answers.

4.5 Cloud Infrastructure

Hosting: AWS EC2, Google Cloud Run, or DigitalOcean **Storage:** AWS S3 or Google Cloud Storage for video files **Queue Processing:** Celery + Redis for handling long-running tasks

5. Development Plan & Timeline

Phase	Task	Duration
Phase 1	System Architecture & Planning	2 weeks
Phase 2	AI Model Development and Integration	3 weeks
Phase 3	Backend & Frontend	3 weeks
Phase 4	Testing & Optimization	2 weeks
Phase 5	Deployment & Scaling	

6. Challenges & Solutions

6.1 Maintaining High Translation Accuracy

- Solution: Fine-tune AI models with academic datasets.

6.2 Cost of Processing Large Videos

- Solution: Optimize efficiency with caching and tiered processing.

6.3 Natural Voice Dubbing

- Solution: Use high-quality TTS models and allow users to select different voice styles.

6.4 AI Chatbot Relevance

- Solution: Implement RAG-based AI to retrieve and generate highly relevant answers

7. Conclusion

Dubjly is an innovative AI-powered platform designed to bridge the language gap for students, providing translated and dubbed educational content along with interactive learning features.

With the right technical execution, it has the potential to revolutionize learning accessibility for Arabic-speaking students.

Next Steps

- 1-Finalize MVP development.
 - 2-Conduct testing with target users.
 - 3-Optimize AI models for improved accuracy.
 - 4- Launch beta version and iterate based on feedback.
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